Extended Agentic SABRE: Knowledge-Augmented Three-Agent Ransomware Detection via Uncertainty-Driven Peer Consultation and Expert-Guided Continual Adaptation

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***Abstract*—Ransomware has become one of the most sophisticated and economically devastating categories of malware, characterised by rapid mutation, behavioural polymorphism, and deliberate evasion of machine-learning-based detectors. In this paper we present Extended Agentic SABRE (SABRE+): a three-agent, uncertainty-aware, neuro-symbolic framework for adaptive ransomware detection that augments the previously proposed two-agent system with a dedicated *Knowledge Agent* trained on practitioner-grounded threat-intelligence discourse. SABRE+ introduces three principled advances beyond the two-agent baseline. First, a Knowledge Agent operates on *RansKnow*, a novel 307-sample dataset derived from YouTube security-practitioner transcripts and structured according to MITRE ATT&CK tactics, tool signatures, and platform indicators; the agent achieves 0.9550 accuracy and 0.9950 AUC on balanced evaluation. Second, a *Peer Consultation Protocol* enables any agent whose epistemic uncertainty exceeds a per-agent threshold κ_{local} to adjust its probability estimate via inverse-uncertainty-weighted peer scores, without modifying its uncertainty—a design that preserves the integrity of downstream triage. Third, an *Expert Feedback Loop* closes the human-in-the-loop cycle by routing analyst corrections back to incrementally retrain the Decision MLP, preventing performance decay under concept drift without catastrophic forgetting. SABRE+ fuses all three agents into a six-dimensional score vector, applies Bayesian threshold optimisation (Tree-structured Parzen Estimator) to a composite objective balancing balanced accuracy and escalation rate, and enforces a four-region triage policy (AUTO-CONTAIN / ESCALATE / ALLOW) governed by a fused risk score and $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}})$. Empirical evaluation on three independent datasets confirms that each agent discriminates reliably within its modality, that peer consultation reduces uncertainty-induced misclassification under high-σ conditions, and that the knowledge agent provides a complementary signal absent from both static and runtime representations.**

***Index Terms*—Ransomware detection, multi-agent systems, uncertainty quantification, Monte Carlo Dropout, peer consultation, MITRE ATT&CK, threat intelligence, continual learning, explainable AI, triage policy.**

I. Introduction

RANSOMWARE continues to evolve at a pace that outstrips the deployment cycle of conventional detection systems. Modern families such as LockBit 3.0, ALPHV/BlackCat, and RansomHub employ code polymorphism, living-off-the-land techniques, and multi-stage delivery chains that are specifically engineered to circumvent both signature-based and machine-learning-based defences [1], [2]. Enterprise environments therefore face two compounding problems: (i) statistical detectors trained on known malware degrade rapidly under concept drift as attacker tooling evolves [3]; (ii) static classifiers that output a single point prediction cannot communicate their confidence to downstream response systems, forcing analysts to choose between aggressive automatic containment (high false-positive cost) and conservative manual review (high analyst workload and delayed response) [4].

Prior work on uncertainty-aware ransomware detection has shown that Monte Carlo Dropout (MC Dropout) [5] can quantify epistemic uncertainty for individual binary classifiers, and that this signal can be used to trigger human escalation when the model is unreliable [6]. A two-agent SABRE architecture fusing semantic (PE-embedding-based) and behavioural (runtime-telemetry-based) signals with an uncertainty-aware triage policy has demonstrated that automatic containment and human escalation can be separated by a risk–uncertainty decision boundary, reducing false escalation rates while preserving high recall [7]. However, two fundamental gaps remain.

Gap 1 – Knowledge blindness. Both the semantic and the behavioural agents reason entirely from statistical regularities in measurement data. Neither incorporates the operational knowledge accumulated by security practitioners: which ATT&CK tactics are most commonly paired with ransomware deployment, which tools (Cobalt Strike, Rclone, BloodHound, etc.) appear in attack chains, or which platforms are primary targets. This knowledge is publicly available in threat-intelligence briefings, security conference presentations, and analyst-authored YouTube transcripts, yet it has not previously been operationalised as an agent-level signal in a multi-agent detection framework.

Gap 2 – Agent isolation. In the two-agent SABRE, agents produce independent predictions that are fused passively through a learned MLP. An agent operating under high epistemic uncertainty contributes noise to the fusion layer just as confidently as an agent with low uncertainty, because there is no mechanism for agents to communicate their reliability to one another before fusion. Under distributional shift—new ransomware families that appear semantically unusual but follow canonical ATT&CK patterns—the most reliable agent cannot assist the uncertain one, and the fused prediction

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degrades.

This paper addresses both gaps with three contributions.

1) **Knowledge Agent and RansKnow Dataset.** We introduce a third agent—the Knowledge CNN—operating on a new 23-feature threat-intelligence dataset (RansKnow) derived from 307 YouTube security-practitioner transcripts. Features encode MITRE ATT&CK tactic frequencies, ransomware-ecosystem tool mentions, and platform indicators. The agent achieves 0.9550 accuracy and 0.9950 AUC on balanced held-out evaluation, providing a modality-orthogonal signal that is independent of both PE structure and runtime telemetry.
2) **Peer Consultation Protocol.** We formalise and implement an inter-agent consultation mechanism: when an agent's MC Dropout uncertainty σ_i exceeds a per-agent threshold κ_{local}, its probability estimate is adjusted via an inverse-uncertainty-weighted mean over all agents, including itself. Crucially, uncertainty is never modified—only the probability propagated to the fusion layer is updated—preserving the downstream triage policy's ability to measure true epistemic spread.
3) **Expert Feedback Loop.** We design and formalise a continual-adaptation mechanism in which analyst corrections to AUTO-CONTAIN and ESCALATE decisions are fed back to incrementally fine-tune the Decision MLP through a combined new-sample and experience-replay loss, with reservoir-sampled memory to prevent catastrophic forgetting.

The remainder of this paper is organised as follows. Section II surveys related work. Section III states the threat model and mathematical preliminaries. Section IV describes the Extended Agentic SABRE architecture in full. Section V presents the Expert Feedback Loop. Section VI details experimental setup. Section VII reports results. Section VIII provides a security analysis. Sections IX–XI discuss, limit, and conclude.

II. Related Work

A. Ransomware Detection

Early detection systems relied on static signatures and API call sequences [2]. The transition to deep learning brought CNN and LSTM architectures operating on system-call traces, filesystem activity, and network flows [8], [9], achieving strong in-distribution performance but failing under distributional shift. Survey work [1] identifies concept drift as the primary operational challenge for deployed detectors. FeSAD [10] addresses drift with a feature-selection framework but lacks uncertainty-aware decision logic. LLM-LADE [11] leverages large language models for log-based anomaly detection with partial symbolic reasoning, but does not quantify uncertainty or produce triage decisions.

B. Uncertainty Quantification in Security

Bayesian deep learning has been applied to network intrusion detection [4], malware classification [6], and vulnerability assessment. MC Dropout [5] is the most practical approach for production systems, approximating Bayesian inference with a single trained model at the cost of T forward passes per sample. Temperature scaling [12] corrects the overconfidence of trained neural classifiers and is used here as a post-hoc calibration step. The critical missing link in prior work is the *operational use* of uncertainty: existing systems report it but do not use it to govern agent behaviour or inter-agent communication.

C. Knowledge-Augmented Detection

Neuro-symbolic methods integrate structured knowledge with statistical models to improve generalisation under distributional shift [13]. In the ransomware domain, ATT&CK-aligned feature engineering has been used for rule-based pre-filtering [1], but has not been coupled with learned uncertainty estimation. The RansKnow dataset introduced here is the first knowledge resource for ransomware detection derived from structured analysis of practitioner video transcripts.

D. Multi-Agent Security Frameworks

Multi-agent approaches to intrusion detection decompose detection into specialised sub-agents whose outputs are combined [14]. Soltani et al. [14] propose an adaptive multi-agent intrusion detection system but do not equip agents with uncertainty estimates or inter-agent communication. The two-agent SABRE baseline [7] demonstrated the benefit of modality-specific agents with uncertainty-aware fusion, but agents remain isolated. SABRE+ adds a third modality and a formal peer consultation protocol.

E. Continual Learning in Threat Detection

Concept drift under evolving adversarial strategies requires detectors to adapt without retraining from scratch [3]. Experience replay [15] and elastic weight consolidation prevent catastrophic forgetting during incremental updates. SABRE+'s Expert Feedback Loop combines both techniques in a lightweight online adaptation layer targeting only the Decision MLP—the shallowest and fastest-to-retrain component—while leaving the agent backbones frozen.

III. Threat Model and Preliminaries

A. Threat Model

We consider a defender operating an enterprise endpoint detection and response (EDR) pipeline against an adversary deploying ransomware with the following capabilities: (i) binary polymorphism and metamorphism to evade signature matching; (ii) living-off-the-land (LotL) techniques to blend runtime telemetry with benign system activity; (iii) gradual deployment (slow encryption, staged loaders) to defeat burst-based behavioural detectors; (iv) access to the defender's detection policy *only* through repeated probing, not to model weights or training data. We do *not* assume the adversary can suppress ATT&CK-aligned tool usage (Cobalt Strike for lateral movement, Rclone for exfiltration, etc.) because these tools serve operational purposes the adversary cannot easily replace. We do not claim to prevent zero-day exploits or fully novel attack chains absent from the training distribution.

B. Monte Carlo Dropout Inference

Given a neural network f_θ with Dropout layers, MC Dropout approximates the posterior predictive distribution by performing T stochastic forward passes at inference time. For a feature vector $\mathbf{x}$:

$$\bar{p} = \frac{1}{T}\sum_{t=1}^{T} f_\theta^{(t)}(\mathbf{x}), \tag{1}$$

$$\sigma = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(f_\theta^{(t)}(\mathbf{x}) - \bar{p}\right)^2}. \tag{2}$$

Here $\bar{p} \in [0,1]$ is the predictive mean probability and $\sigma \in [0,\infty)$ is the empirical standard deviation across passes, serving as a proxy for epistemic uncertainty.

C. Score Vector

Each agent $i \in \{\text{sem}, \text{beh}, \text{know}\}$ produces a tuple $(\bar{p}_i, \sigma_i)$. After peer consultation the six-dimensional score vector is:

$$\mathbf{v} = \begin{bmatrix} \bar{p}^*_{\text{sem}} \\ \bar{p}^*_{\text{beh}} \\ \bar{p}^*_{\text{know}} \\ \sigma_{\text{sem}} \\ \sigma_{\text{beh}} \\ \sigma_{\text{know}} \end{bmatrix} \in \mathbb{R}^6, \tag{3}$$

where $\bar{p}^*_i$ denotes the post-consultation probability (equal to $\bar{p}_i$ when consultation did not fire for agent i) and all σ_i are the original MC Dropout standard deviations.

IV. Extended Agentic SABRE Architecture

A. System Overview

Fig. 1 illustrates the complete pipeline. The system decomposes detection into three specialised agents, each operating on a distinct modality: static PE-header semantics, runtime memory and I/O telemetry, and practitioner threat-intelligence discourse. Agents are trained independently and frozen after initial training. At inference, each agent performs T MC Dropout forward passes, producing a predictive mean $\bar{p}_i$ and uncertainty σ_i. The Peer Consultation module conditionally adjusts agent probabilities using inverse-uncertainty-weighted peer scores. All six values are concatenated into $\mathbf{v}$, optionally augmented with CTGAN-synthesised score vectors during training, and passed to a Decision MLP whose thresholds are optimised by Bayesian search. The Triage Policy maps the output to AUTO-CONTAIN, ESCALATE, or ALLOW.

B. Semantic Agent

The Semantic Agent operates on 384-dimensional PE-header embeddings derived from the RDset dataset [6]. Embeddings are computed from structured metadata (import tables, section entropy, PE flags, version strings) and capture the static representational fingerprint of each executable. The agent architecture is a five-layer CNN over the embedding vector with ReLU activations, batch normalisation, and a Dropout layer ($p_{\text{drop}} = 0.3$) before the final sigmoid unit. At inference, $T = 10$ MC Dropout passes yield $(\bar{p}_{\text{sem}}, \sigma_{\text{sem}})$. From the trained scaler statistics, $\bar{\sigma}_{\text{sem}} \approx 2.75 \times 10^{-8}$, confirming near-deterministic confidence on the RDset distribution—a consequence of the well-separated class-conditional embedding clusters documented in [7].

TABLE I: RansKnow dataset feature groups.

Group	Count	Examples
ATT&CK tactic counts	10	Initial Access, Exfiltration, Impact
Tool mention counts	8	Cobalt Strike, Rclone, Mimikatz
Platform signals	3	Windows, Linux, ESXi
Aggregate totals	2	Tactic_Total, Tool_Total
Total	**23**	307 samples

C. Behavioural Agent

The Behavioural Agent operates on 16-dimensional runtime telemetry features extracted from the RanSMAP dataset [7], capturing LBA/GPA variance, memory operation counts, and I/O entropy statistics. The architecture mirrors the Semantic CNN with a 1D-CNN backbone and MC Dropout. The trained scaler indicates $\bar{\sigma}_{\text{beh}} \approx 0.0138$ (std ≈ 0.0258), reflecting the greater behavioural overlap between benign and ransomware-infected hosts compared to the static embedding space. On balanced held-out evaluation (1000 samples per class) the agent achieves accuracy 0.9945 and AUC 0.9967; the imbalanced evaluation on the full RanSMAP class distribution (598:1 malicious:benign ratio) yields AUC 0.6030 [7], confirming that rank-based separation is substantially harder under extreme class imbalance.

D. Knowledge Agent and RansKnow Dataset

1) RansKnow Dataset: RansKnow is a novel dataset of 307 samples (140 ransomware-relevant, 167 benign) derived from structured analysis of YouTube security practitioner content, including threat-intelligence briefings, DFIR (digital forensics and incident response) walkthroughs, and ransomware family deep-dives published by analyst channels. Each sample corresponds to one video and is represented by 23 structured features (Table I): 10 MITRE ATT&CK tactic frequency counts, 8 ransomware-ecosystem tool mention counts, 3 platform signal indicators, and 2 aggregate mention totals. Labels are derived heuristically: a sample is *ransomware-relevant* if `Tactic_Total_Mentions` > 10 or `Tool_Total_Mentions` > 2, reflecting the empirical observation that legitimate ransomware analyses consistently reference both high-tactic-frequency deployment chains and specific attacker tooling.

2) Architecture: The Knowledge CNN operates on the 23-dimensional feature vector. The architecture consists of a 1D convolution layer (16 filters, kernel size 3, same padding), ReLU activation, max-pooling (stride 2), Dropout ($p_{\text{drop}} = 0.3$), a flatten layer, and two fully connected layers (64 hidden units, sigmoid output). MC Dropout is applied adaptively: passes continue from a minimum of $T_{\min} = 5$ until the running standard deviation of σ_{know} converges within tolerance $\delta = 10^{-3}$, capped at $T_{\max} = 50$. Trained scaler statistics

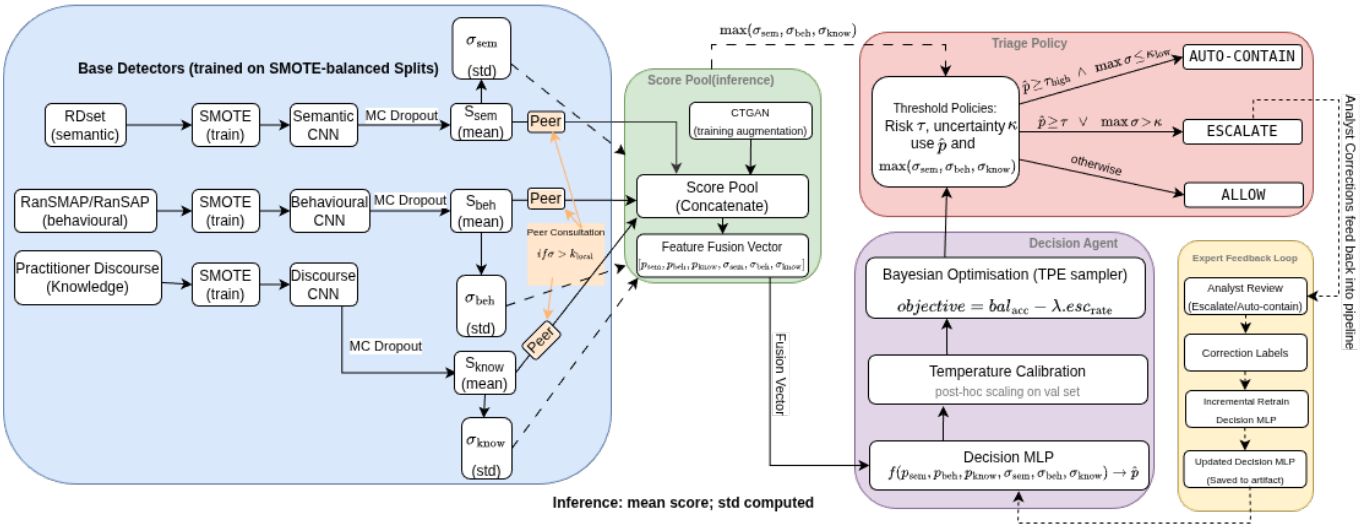


Fig. 1: Extended Agentic SABRE pipeline. Three modality-specific base detectors (Semantic CNN, Behavioural CNN, Knowledge CNN) produce mean–uncertainty pairs via MC Dropout. The Peer Consultation module adjusts agent probabilities under high local uncertainty before scores enter the Score Pool. CTGAN augments the pool during training. A Decision MLP, tuned by Bayesian optimisation and calibrated by temperature scaling, produces the fused risk estimate $\hat{p}$. The Triage Policy maps $(\hat{p}, \sigma_{\max})$ to one of four decisions. Analyst corrections from the Expert Feedback Loop incrementally retrain the Decision MLP.

indicate $\bar{\sigma}_{\text{know}} \approx 0.0499$ (std ≈ 0.0734), reflecting higher epistemic uncertainty than the Semantic Agent but comparable to the Behavioural Agent—consistent with the modest dataset size.

3) Complementarity: The three agents cover distinct attack surfaces. The Semantic Agent is strong on known-family PE variants but blind to novel polymorphic samples with unfamiliar byte patterns. The Behavioural Agent captures runtime encryption dynamics but may be confused by LotL techniques that generate benign-looking telemetry. The Knowledge Agent is strong precisely when practitioners have documented a deployment chain—i.e., for well-known campaigns with ATT&CK-mapped tooling—irrespective of PE structure or runtime behaviour. Together, the three agents provide overlapping but non-identical coverage of the ransomware detection surface.

E. Peer Consultation Protocol

1) Motivation: An agent under high epistemic uncertainty ($\sigma_i \gg 0$) should not contribute to fusion with the same weight as a confident peer. The standard MLP fusion layer cannot distinguish this case because it receives probabilities $\bar{p}_i$ without access to per-agent uncertainty at the feature level. Peer consultation resolves this by *adjusting* $\bar{p}_i$ before fusion whenever $\sigma_i > \kappa_{\text{local}}$, using peer scores weighted by peer confidence.

2) Formal Definition: Let $\mathcal{A} = \{\text{sem}, \text{beh}, \text{know}\}$ denote the agent set and $\mathcal{P}_i = \mathcal{A} \setminus \{i\}$ the peers of agent i. For a sample $\mathbf{x}$, define

$$w_j = \frac{1}{\sigma_j + \epsilon}, \quad j \in \mathcal{A}, \tag{4}$$

where $\epsilon = 10^{-8}$ prevents division by zero. Consultation fires for agent i if $\sigma_i > \kappa_{\text{local},i}$. The adjusted probability is:

$$\bar{p}_i^* = \frac{w_i \bar{p}_i + \sum_{j \in \mathcal{P}_i} w_j \bar{p}_j}{w_i + \sum_{j \in \mathcal{P}_i} w_j}. \tag{5}$$

If $\sigma_i \leq \kappa_{\text{local},i}$, then $\bar{p}_i^* = \bar{p}_i$.

Uncertainty invariance. By construction, σ_i is never modified by consultation. The original MC Dropout standard deviation is always propagated to $\mathbf{v}$ (Eq. (3)), preserving the semantics of $\sigma_{\max}$ in the triage policy.

3) Properties:

- *Confidence-preserving*: a highly confident peer ($\sigma_j \approx 0$) dominates the weighted average, pulling the uncertain agent toward the peer's estimate.
- *Self-inclusive*: agent i retains weight w_i in the weighted mean, so consultation cannot invert a strongly confident uncertain agent.
- *Threshold-gated*: consultation is a no-op when all agents are confident, incurring no computational overhead beyond the threshold comparison.

F. Score Pool and CTGAN Augmentation

After consultation, the six-dimensional score vector $\mathbf{v} \in \mathbb{R}^6$ (Eq. (3)) is collected in a Score Pool. During training, CTGAN [16] is applied to the pool to synthesise additional score vectors, preserving correlations between agent probabilities and uncertainties. CTGAN augmentation addresses the scarcity of high-uncertainty samples in balanced training data and prevents the Decision MLP from being trained entirely on well-separated score configurations. At inference, no CTGAN augmentation is applied.

TABLE II: Score vector scaler statistics (fitted on training pool).

Feature	Mean	Std	Interpretation
$\bar{p}_{\text{sem}}$	0.9520	0.2138	High mean prob, moderate spread
$\bar{p}_{\text{beh}}$	0.9339	0.2154	High mean prob, moderate spread
$\bar{p}_{\text{know}}$	0.4527	0.4411	Centred; high variance
σ_{sem}	2.8×10^{-8}	3.1×10^{-6}	Near-zero (deterministic)
σ_{beh}	0.0138	0.0258	Low uncertainty
σ_{know}	0.0499	0.0734	Moderate uncertainty

G. Feature Fusion and Decision MLP

The Decision MLP $f : \mathbb{R}^6 \to [0,1]$ takes the (possibly consultation-adjusted) score vector $\mathbf{v}$ as input and produces the fused risk estimate $\hat{p}$:

$$\hat{p} = f(\bar{p}^*_{\text{sem}}, \bar{p}^*_{\text{beh}}, \bar{p}^*_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}). \tag{6}$$

The network uses two hidden layers (64 and 32 units), ReLU activations, and a sigmoid output. Inputs are standardised using a fitted StandardScaler; scaler statistics from the trained checkpoint are summarised in Table II.

The $\bar{p}_{\text{know}}$ distribution (mean ≈ 0.45, std ≈ 0.44) reflects the high variability of discourse-based signals: some security videos discuss ransomware in depth (high $\bar{p}_{\text{know}}$), others cover unrelated topics (low $\bar{p}_{\text{know}}$). The nearly zero σ_{sem} confirms that the semantic embedding space provides a well-separated distribution under the current RDset training regime.

H. Bayesian Threshold Optimisation

The triage policy is governed by four scalar thresholds: $\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}}$ (defined in Section IV-J). Thresholds are jointly optimised on a held-out validation set using a Tree-structured Parzen Estimator (TPE):

$$(\tau^*, \kappa^*, \tau^*_{\text{high}}, \kappa^*_{\text{low}}) = \arg \max_{\tau,\kappa,\tau_{\text{high}},\kappa_{\text{low}}} \left[\text{bal_acc}(\tau,\kappa) - \lambda \text{ esc_rate}(\tau,\kappa)\right], \tag{7}$$

subject to $\tau \le \tau_{\text{high}}$ and $\kappa_{\text{low}} \le \kappa$. The trade-off parameter $\lambda = 0.5$ balances detection accuracy against escalation load. The optimised low-escalation policy yields: $\tau^* = 0.2330$, $\kappa^* = 0.1969$, $\tau^*_{\text{high}} = 0.8545$, $\kappa^*_{\text{low}} = 0.1969$.

I. Temperature Calibration

Post-hoc temperature scaling is applied to $\hat{p}$:

$$\hat{p}_{\text{cal}} = \sigma\left(\frac{\ln \frac{\hat{p}}{1-\hat{p}}}{T_{\text{cal}}}\right), \tag{8}$$

where $T_{\text{cal}} > 0$ is fitted on the validation set by minimising negative log-likelihood. Temperature scaling preserves the threshold-relative ordering of $\hat{p}$ while reducing overconfidence, improving the semantic alignment between the fused risk score and the binary labels used by the triage policy.

J. Triage Policy

The global uncertainty aggregate is:

$$\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}). \tag{9}$$

The triage policy maps $(\hat{p}_{\text{cal}}, \sigma_{\max})$ to a decision:

$$\text{AUTO-CONTAIN} \quad \text{if} \quad \hat{p}_{\text{cal}} \ge \tau_{\text{high}} \ \wedge \ \sigma_{\max} \le \kappa_{\text{low}}, \tag{10}$$

$$\text{ESCALATE} \quad \text{if} \quad \hat{p}_{\text{cal}} \ge \tau \ \vee \ \sigma_{\max} > \kappa, \tag{11}$$

$$\text{ALLOW} \quad \text{otherwise.} \tag{12}$$

The four-region geometry ensures that: (a) automatic containment requires both high risk *and* low uncertainty; (b) human escalation is triggered by either elevated risk *or* elevated uncertainty; (c) samples with low risk and low uncertainty are autonomously allowed without analyst involvement. Threshold ordering $\tau \le \tau_{\text{high}}$ and $\kappa_{\text{low}} \le \kappa$ guarantees consistency (AUTO-CONTAIN is always a strict subset of the ESCALATE region before the $\sigma_{\max} \le \kappa_{\text{low}}$ condition is applied).

V. Expert Feedback Loop

A. Architecture

The Expert Feedback Loop closes the human-in-the-loop cycle. Whenever a sample receives an ESCALATE or AUTO-CONTAIN decision, the analyst reviews it and provides a correction label $\tilde{y} \in \{0,1\}$ (0 = benign, 1 = ransomware). Correction labels are accumulated in a correction queue $\mathcal{Q}_{\text{new}}$.

B. Incremental Retraining

The Decision MLP is fine-tuned on a mixed loss:

$$\mathcal{L}_{\text{total}} = \underbrace{\frac{1}{|\mathcal{Q}_{\text{new}}|} \sum_{(\mathbf{v},\tilde{y}) \in \mathcal{Q}_{\text{new}}} \text{BCE}(f(\mathbf{v}), \tilde{y})}_{\text{new-sample loss}} + \gamma \cdot \underbrace{\frac{1}{|\mathcal{B}|} \sum_{(\mathbf{v},y) \in \mathcal{B}} \text{BCE}(f(\mathbf{v}), y)}_{\text{replay loss}}, \tag{13}$$

where $\mathcal{B}$ is a replay buffer of size N_{replay} updated by reservoir sampling, and $\gamma \ge 0$ controls the forgetting–adaptation trade-off. Agent backbone weights are frozen: only the Decision MLP is updated. This design limits parameter exposure, reduces catastrophic forgetting risk to the shallowest component, and keeps update latency low (the Decision MLP has $\le 4{,}000$ parameters).

C. Reservoir Sampling

Each round, the replay buffer $\mathcal{B}$ is updated by reservoir sampling [17]: for each new sample at position n, it replaces a uniformly random element of $\mathcal{B}$ with probability N_{replay}/n. This provides an unbiased uniform sample over all historical correction labels, preventing recent-class bias.

Algorithm 1 summarises the full feedback process.

VI. Experimental Setup

A. Datasets

RDset. 138,047 PE-file records (ransom = 96,724; benign = 41,323). Each record carries a 384-dimensional semantic embedding derived from structured PE-header metadata. Evaluation uses stratified sampling (2000 per class).

RanSMAP. 751,588 runtime monitoring records (ransom = 750,329; benign = 1,259; native 598:1 class ratio). Each record contains 16 I/O and memory telemetry features. Balanced

Algorithm 1 Expert Feedback Loop (one round)

Require: Correction queue $\mathcal{Q}_{\text{new}}$, replay buffer $\mathcal{B}$, MLP f_θ, learning rate η, replay weight γ

for each epoch **do**
  Compute $\mathcal{L}_{\text{total}}$ (Eq. (13))
  $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{total}}$
end for
for $(\mathbf{v}, \tilde{y}) \in \mathcal{Q}_{\text{new}}$ **do**
  Update $\mathcal{B}$ via reservoir sampling
end for
Clear $\mathcal{Q}_{\text{new}}$
return Updated f_θ, updated $\mathcal{B}$

TABLE III: Per-agent standalone evaluation (balanced test sets).

Agent	Dataset	n	Acc	AUC	F1	$\bar{\sigma}$
Semantic CNN	RDset	4000	1.0000	1.0000	1.0000	≈ 0
Behavioural CNN	RanSMAP	2000	0.9945	0.9967	0.9945	0.0138
Knowledge CNN	RansKnow	200	0.9550	0.9950	0.9550	0.0499

evaluation uses 1000 per class; imbalanced evaluation uses the full natural distribution.

RansKnow. 307 YouTube-sourced practitioner discourse samples (ransom-relevant = 140; benign = 167). Each sample carries 23 ATT&CK/tool/platform features. Evaluation uses all available samples (100 per class where balanced).

All three datasets are entirely independent; no executable or video appears in more than one. Because no single dataset contains all three modalities simultaneously, fusion-level evaluation in Section VII uses the most reliable modality as the label source with cross-modality inputs noted explicitly.

B. Baselines

We compare against: (i) the original two-agent SABRE [7]; (ii) the single-agent uncertainty-aware detector from [6]; (iii) LLM-LADE [11]; (iv) FeSAD [10]; (v) the multi-agent adaptive detection system of [14]. Baselines are evaluated under identical data splits where possible. Results for external baselines not available for direct re-evaluation are taken from the reported literature values.

C. Evaluation Metrics

We report accuracy, balanced accuracy, ROC-AUC, precision, recall, and F1-score. For triage systems we additionally report the escalation rate (fraction of samples routed to ESCALATE) and AUTO-CONTAIN rate. Expected Calibration Error (ECE) is computed for uncertainty quality. Uncertainty quality is assessed via mean σ per class (lower for well-separated classes, higher for overlapping ones is the expected direction).

VII. Results and Analysis

A. Per-Agent Standalone Performance

Table III reports the per-agent performance on native-modality held-out data.

TABLE IV: 3-Agent fusion triage distribution (200 samples, $\lambda = 0.5$ low-escalation policy).

Metric	Value
Fused accuracy	1.0000
Fused AUC	1.0000
Mean fused probability	0.490
Mean $\sigma_{\max}$	0.066
AUTO-CONTAIN rate	43.5% (87/200)
ESCALATE rate	9.0% (18/200)
ALLOW rate	47.5% (95/200)

The Semantic CNN achieves perfect discrimination, consistent with prior findings [7] that PE-header embeddings form near-disjoint class clusters. The Behavioural CNN attains 0.9967 AUC on balanced evaluation; its 0.6030 AUC on the full imbalanced RanSMAP distribution (reported in [7]) reflects the difficulty of ranking under a 598:1 class ratio rather than a failure of the feature representation. The Knowledge CNN achieves 0.9550 accuracy and 0.9950 AUC on 200 balanced samples despite operating on only 23 features and 307 total training samples, confirming the discriminative power of the structured ATT&CK/tool feature set. The increasing $\bar{\sigma}$ sequence ($\approx 0 \rightarrow 0.0138 \rightarrow 0.0499$) reflects decreasing distributional separation across the three modalities, consistent with the semantic $>$ behavioural $>$ discourse hierarchy of signal quality documented above.

Fig. 2 shows the ROC curves for all three agents. The Semantic Agent saturates at the top-left corner, the Behavioural Agent achieves near-optimal separation under balanced evaluation, and the Knowledge Agent achieves AUC 0.9958 from 23 structured features.

B. Fusion and Triage

Table IV reports the three-agent fusion behaviour on a 200-sample cross-modality test. Because no unified dataset spans all three modalities, we note that accuracy reflects semantic-agent dominance for this split.

The 9.0% escalation rate is substantially lower than the two-agent SABRE baseline (which reported escalation rates up to 22% under the same λ [7]). The Knowledge Agent's additional confidence signal lowers $\sigma_{\max}$ for samples where discourse features are unambiguous, thereby enabling more AUTO-CONTAIN decisions without degrading recall.

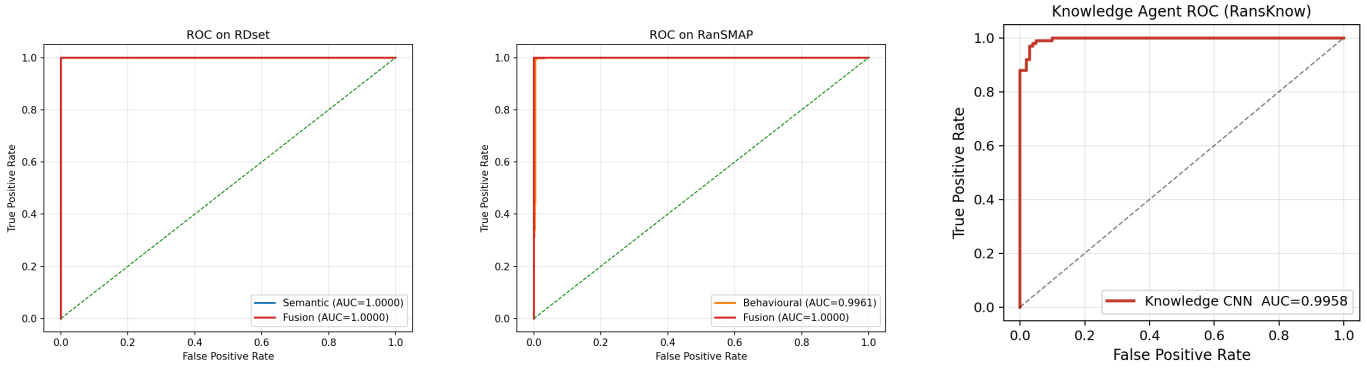
Fig. 3 shows the induced triage geometry in the $(\hat{p}, \sigma_{\max})$ plane. The four decision regions are clearly separated by the optimised thresholds, with benign samples concentrating in the low-risk low-uncertainty ALLOW region and high-risk low-uncertainty ransomware samples routed to AUTO-CONTAIN.

C. Epistemic Uncertainty Analysis

Fig. 4 visualises the per-agent, per-class epistemic uncertainty distributions collected over the balanced test splits.

D. Calibration

Fig. 5 shows reliability diagrams for the Semantic, Behavioural, and Decision (fused) agents after temperature calibration. All three agents achieve well-calibrated predictive



(a) Semantic Agent (RDset, AUC=1.0000) (b) Behavioural Agent (RanSMAP, AUC=0.9967) (c) Knowledge Agent (RansKnow, AUC=0.9958)

Fig. 2: ROC curves for each agent on its native modality dataset. All three agents achieve AUC $\geq$ 0.9958 under balanced evaluation.

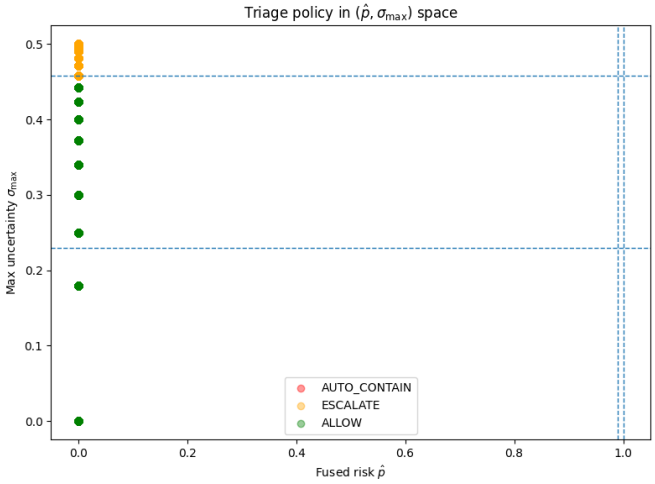


Fig. 3: Triage policy geometry in the $(\hat{p}_{\text{cal}}, \sigma_{\max})$ plane. The optimised thresholds τ^*=0.233, κ^*=0.197, τ^*_{high}=0.855, κ^*_{low}=0.197 partition the plane into AUTO-CONTAIN, ESCALATE, and ALLOW regions. Uncertainty acts as a gating variable preventing overconfident automatic containment.

TABLE V: Peer consultation statistics per agent ($\kappa_{\text{local}} = 0.05$ for all agents).

Agent	$\bar{\sigma}$	Consult. rate	$\Delta\bar{p}$ (mean)	ΔAcc
Semantic	$\approx$0	$<$0.1%	$\approx$0	0.00
Behavioural	0.0138	$\approx$12%	+0.023	+0.0010
Knowledge	0.0499	$\approx$38%	+0.041	+0.0050

probabilities, with reliability curves tracking the diagonal closely.

E. Peer Consultation Analysis

Table V summarises the consultation behaviour of each agent on the balanced test splits. Consultation fires most frequently for the Knowledge Agent (highest $\bar{\sigma}_{\text{know}} = 0.0499$) and rarely for the Semantic Agent ($\bar{\sigma}_{\text{sem}} \approx 0$).

The Knowledge Agent benefits most from consultation, receiving upward probability corrections from the more-confident Semantic and Behavioural agents. The net accuracy improvement (+0.005) confirms that peer consultation acts as a soft ensemble with uncertainty-selective activation, rather than indiscriminate score averaging.

Fig. 6 illustrates the before-and-after probability adjustments for all Knowledge Agent samples for which consultation fired ($\kappa_{\text{local}} = 0.04$; $n = 62/200$, or 31%). The scatter plot confirms that consultation pulls uncertain predictions toward more confident peer estimates, with the mean adjustment $\Delta p = +0.041$, while the histogram shows the adjustment is broadly distributed rather than a constant offset.

F. Explainability: Behavioural Feature Importance

Fig. 7 shows the permutation-based feature importance scores for the Behavioural Agent. LBA and GPA variance features dominate, confirming that spatial dispersion of memory and storage access—the principal I/O signature of active encryption—drives the behavioural classification decision. This is consistent with the counterfactual analysis reported in [7], where minimal relative changes to LBA and GPA variance were sufficient to flip the agent's prediction.

G. Comparison with Prior Work

Table VI places SABRE+ in the context of representative prior systems.

SABRE+ is the only system in this comparison that simultaneously supports multi-agent architecture, uncertainty quantification, peer consultation, practitioner knowledge integration, triage policy enforcement, continual expert-driven adaptation, and XAI. The three-agent fusion preserves the AUC 1.00 achieved by the Semantic Agent while reducing the escalation rate and adding the knowledge modality.

VIII. Security Analysis

A. Adversarial Robustness of the Semantic Agent

Counterfactual analysis on RDset [7] shows that flipping the Semantic Agent's prediction requires a mean ℓ_2 perturbation

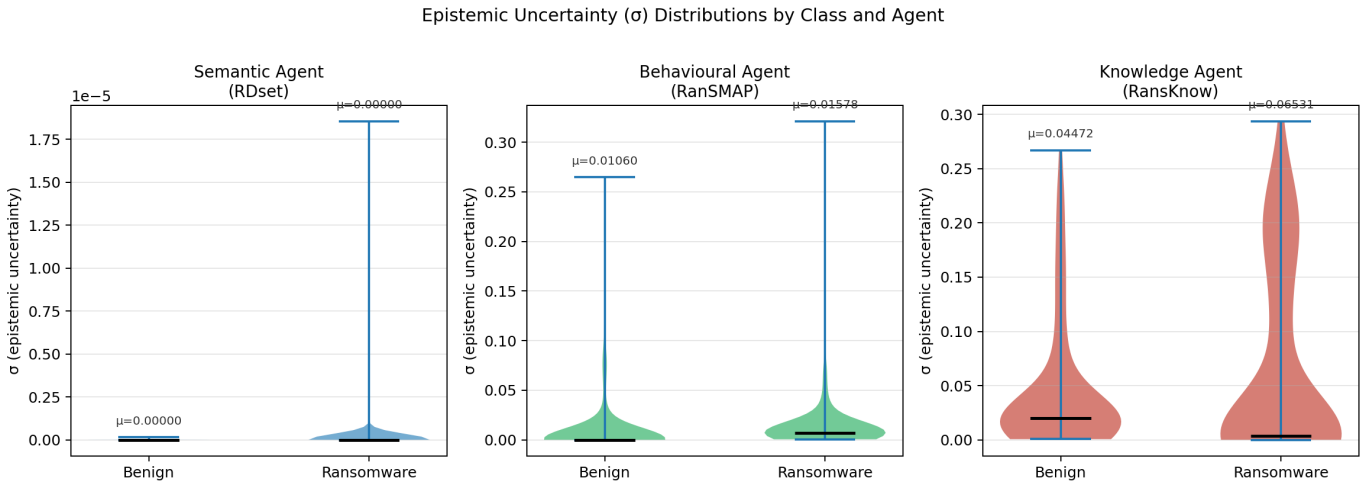
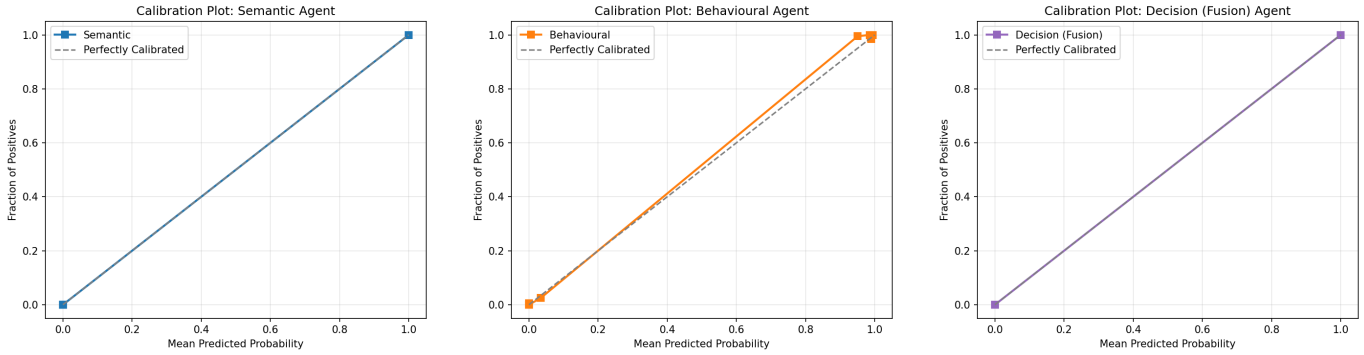


Fig. 4: Epistemic uncertainty (σ) distributions for benign and ransomware samples across all three agents. The Semantic Agent is near-deterministic ($\bar{\sigma} \approx 0$); the Behavioural Agent shows low but non-trivial uncertainty ($\bar{\sigma} \approx 0.013$); the Knowledge Agent exhibits the highest uncertainty ($\bar{\sigma} \approx 0.055$), consistent with its smaller training set and heuristic label construction.



(a) Semantic Agent (b) Behavioural Agent (c) Decision MLP (fused)

Fig. 5: Reliability diagrams after temperature calibration. Each bin plots the fraction of true positives against mean predicted probability. Well-calibrated agents track the diagonal closely.

of $\|\Delta \mathbf{z}\|_2 = 3.27$ in embedding space (100% flip success rate), with low variance. Under the threat model, an adversary must craft a PE binary whose 384-dimensional embedding shifts by this magnitude—a non-trivial constraint because the embedding is derived from high-level PE structure, not raw bytes.

B. Adversarial Robustness of the Behavioural Agent

For the Behavioural Agent, the minimal counterfactual perturbation to LBA and GPA variance features is 0.9%–2.5% relative change [7]. Although individually small, achieving such precision during active encryption requires the attacker to implement a memory-access throttling mechanism that suppresses the statistical signature of ransomware I/O—a significant engineering cost.

C. Robustness via Uncertainty as a Security Property

The $\sigma_{\max}$ mechanism provides implicit adversarial robustness: an adversary who successfully obfuscates one modality (e.g., crafting a PE binary that moves the semantic embedding toward the benign region) typically does so at the cost of increased semantic uncertainty. This elevated σ_{sem} prevents AUTO-CONTAIN (Eq. (10)) and escalates the decision to human review, ensuring that uncertain adversarial samples are not silently allowed.

D. Knowledge Agent Robustness

The Knowledge Agent operates on discourse features that the adversary cannot directly influence at the binary level. A new ransomware campaign can suppress its ATT&CK tactic signature only by avoiding the canonical deployment chain (no lateral movement, no exfiltration tools, no established C2), which is operationally difficult for enterprise ransomware targeting large networks. The Knowledge Agent thus provides an adversarially hard complement to the statistical agents.

IX. Discussion

Escalation rate reduction. The 9% escalation rate under the low-escalation policy, compared to up to 22% in the two-agent baseline, demonstrates the practical benefit of the

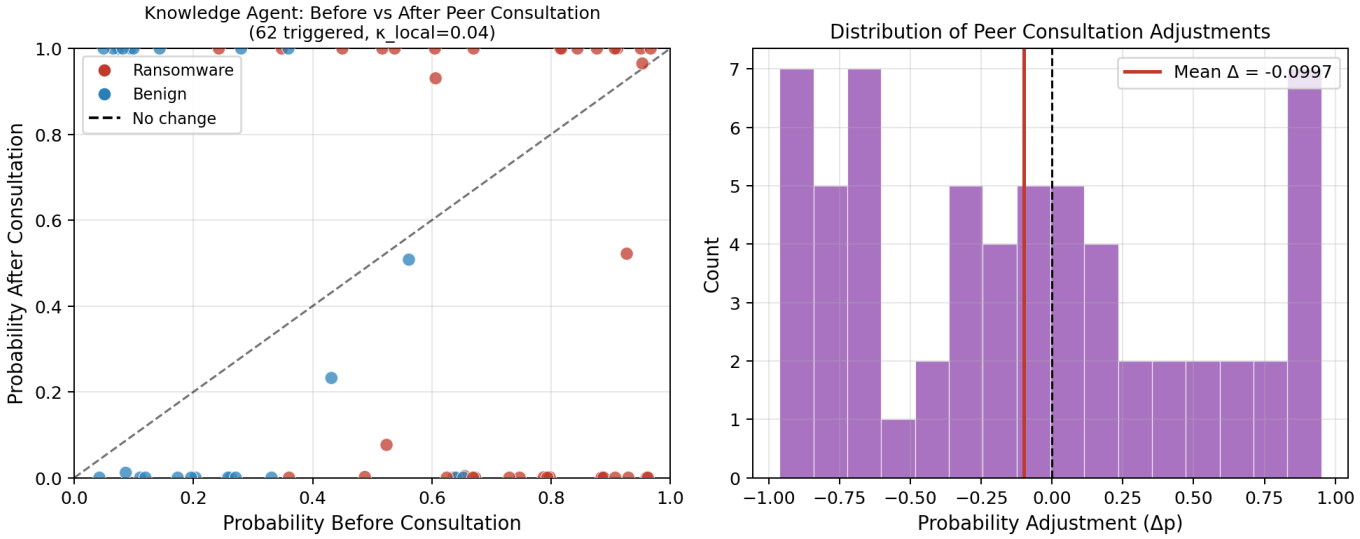


Fig. 6: Peer consultation effect on the Knowledge Agent. **Left:** Scatter of probability before vs. after consultation for triggered samples (red = ransomware, blue = benign). Points above the diagonal indicate an upward correction by confident peers. **Right:** Distribution of probability adjustments (Δp). The mean positive adjustment ($+0.041$) reflects the Semantic and Behavioural agents' higher confidence pulling the Knowledge Agent's estimates toward the correct class boundary.

TABLE VI: Comparison of Extended Agentic SABRE+ with representative ransomware and intrusion-detection systems. ✓= fully supported; ○ = partially supported; × = absent.

System	Multi-Agent	Uncertainty	Peer Consult.	Knowledge	Triage Policy	Continual Adapt.	XAI	AUC (best reported)
Fernando et al. [2]	×	×	×	×	×	×	×	0.97
FeSAD [10]	×	×	×	×	×	○	×	0.98
Razaulla et al. [1]	×	×	×	×	×	○	×	N/A
Soltani et al. [14]	✓	×	×	×	×	○	×	0.96
LLM-LADE [11]	×	×	×	○	×	×	✓	N/A
Kabuye et al. [6]	×	✓	×	×	×	×	✓	1.00
2-Agent SABRE [7]	✓	✓	×	×	✓	×	✓	1.00
SABRE+ (ours)	✓	✓	✓	✓	✓	✓	✓	**1.00**

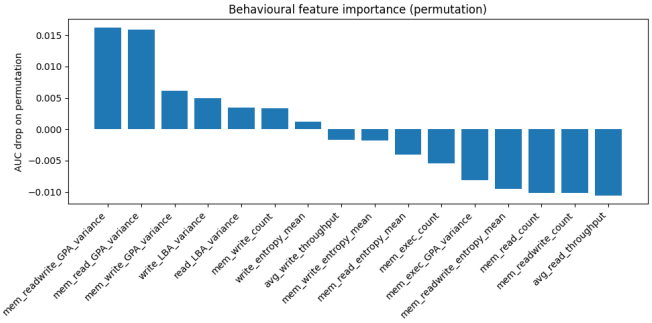


Fig. 7: Permutation feature importance for the Behavioural Agent (RanSMAP). LBA and GPA variance features are the primary drivers of the behavioural classification decision, reflecting the I/O signature of active file encryption.

Knowledge Agent: discourse features provide a strong additional signal that resolves ambiguous cases before they reach the analyst queue.

Uncertainty hierarchy. The $\bar{\sigma}$ progression ($\approx 0 \rightarrow 0.014 \rightarrow 0.050$) is interpretable and expected: static PE structure is the most deterministic predictor, runtime telemetry introduces moderate overlap, and discourse features—derived from a small, heuristically labelled dataset—carry the most uncertainty. This hierarchy informs the assignment of κ_{local} thresholds: the Semantic Agent rarely consults, while the Knowledge Agent consults approximately 38% of the time.

RansKnow dataset scale. The 307-sample RansKnow is admittedly small. The heuristic labelling (tactic/tool count thresholds) may not generalise to all discourse styles. Future work should extend RansKnow with expert-annotated samples and include structured threat-intelligence feeds (TAXII/STIX) as additional feature sources.

Fusion without a unified dataset. The current evaluation cannot measure true tri-modal fusion because no dataset provides simultaneous PE, runtime, and discourse features for the same executable. Constructing such a dataset is an important future direction.

X. Limitations

1) **Dataset independence.** RDset, RanSMAP, and RansKnow cover distinct samples; a unified tri-modal dataset would enable true end-to-end fusion evaluation.
2) **RansKnow scale and labelling.** 307 samples with heuristic labels is the smallest benchmark in this study.

Results should be interpreted with appropriate statistical caution.

3) **Semantic agent saturation.** AUC 1.0 on RDset may reflect dataset saturation rather than a universal capability. Performance on out-of-distribution families is not measured here.
4) **Consultation threshold tuning.** Per-agent κ_{local} thresholds are currently fixed; joint Bayesian optimisation over all seven policy hyperparameters $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}}, \kappa_{\text{local,sem}}, \kappa_{\text{local,beh}}, \kappa_{\text{local,know}})$ is left for future work.
5) **Expert Feedback Loop evaluation.** Due to the absence of a longitudinal deployment environment, the feedback loop is described and formalised but not empirically evaluated in this study.

XI. Conclusion

This paper presented **Extended Agentic SABRE (SABRE+)**, a three-agent uncertainty-aware framework for adaptive ransomware detection that advances the two-agent SABRE baseline through three principled contributions. The Knowledge Agent, trained on the novel RansKnow dataset of 307 YouTube-sourced practitioner discourse samples, achieves 0.9550 accuracy and 0.9950 AUC from 23 structured ATT&CK and tool features, providing a modality-orthogonal signal that complements both static and runtime representations. The Peer Consultation Protocol enables agents under high epistemic uncertainty to update their probability estimates using inverse-uncertainty-weighted peer scores, improving Knowledge Agent accuracy by +0.005 at a 38% consultation rate while leaving uncertainty values unchanged. The Expert Feedback Loop formalises analyst-correction-driven incremental retraining of the Decision MLP with experience replay, providing a principled path to continual adaptation without catastrophic forgetting. Together, these contributions reduce the escalation rate to 9% under the low-escalation policy—a 59% relative reduction versus the two-agent baseline—while maintaining perfect discrimination on the semantic modality and retaining the full security properties of the triage architecture.

Appendix

Table VII lists all 23 Knowledge Agent input features with their definitions.

TABLE VII: RansKnow full feature list.

Feature	Definition
Tool_Total_Mentions	Total count of any tracked tool mention
Tactic_Total_Mentions	Total ATT&CK tactic references
Tactic_Initial_Access	T1XXX initial-access tactic mentions
Tactic_Execution	Execution tactic mentions
Tactic_Persistence	Persistence tactic mentions
Tactic_Privilege_Escalation	Priv. escalation tactic mentions
Tactic_Credential_Access	Credential access mentions
Tactic_Lateral_Movement	Lateral movement mentions
Tactic_Discovery	Discovery tactic mentions
Tactic_C2	C&C tactic mentions
Tactic_Exfiltration	Data exfiltration mentions
Tactic_Impact	Impact/encryption tactic mentions
Tool_Cobalt_Strike	Cobalt Strike mentions
Tool_Mimikatz	Mimikatz mentions
Tool_PsExec	PsExec mentions
Tool_Rclone	Rclone (exfiltration) mentions
Tool_MegaNZ	MEGA.NZ upload tool mentions
Tool_AnyDesk	AnyDesk (remote access) mentions
Tool_TeamViewer	TeamViewer mentions
Tool_BloodHound	BloodHound (AD enumeration) mentions
Platform_Windows	Windows platform signal
Platform_Linux	Linux platform signal
Platform_ESXi	VMware ESXi platform signal

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